

CE 3111

STRUCTURAL ANALYSIS AND DESIGN - I FORMULA + THEORY + PROOF / DERIVATION MASTER NOTE

Built for exam revision - Organized by Section A & Section B and the 8-set distribution used in your CE 3111 question-bank analysis. Formula tables, theory answers, proofs/derivations, memory shortcuts and common-error checks are combined in one file.

Made by 2223033

Source basis: uploaded class notes, lecture slides, assignments, question bank (2016-2024) and 2024 Backlog materials.

How to use this master note

- Read the red-tagged HIGH priority derivations first; these are the proof/formula patterns that repeatedly support numerical questions.
- Before a numerical, scan the Master Formula table of that set and the "Common mistake" column.
- For theory answers, write: definition -> 3 to 5 key points -> governing formula/diagram -> one closing line. This structure is already reflected below.
- Sign conventions can vary across books. In this note, the class-note convention is stated where it matters; keep one convention consistently throughout a solution.

8-set map

Section	Set	Core topics	Main derivation/proof targets
A	1	Stability & determinacy; SFD/BMD	Load-shear-moment differential relations
A	2	Plane truss; three-hinged arch	Parabolic arch equation; $M = M_{\text{beam}} - H y$; UDL gives zero BM
A	3	Wind load; earthquake load	Code-formula chain and distribution checks
A	4	Unit-load/virtual work; conjugate beam	$\Delta = \int Mm/EI dx$; $V^* = \theta$, $M^* = \Delta$
B	5	Influence Line Diagram (ILD)	Simple-beam ILDs; Muller-Breslau principle
B	6	Approximate frame analysis	Portal and cantilever assumptions; cantilever force distribution
B	7	Mill bent / bridge portal; space truss	3D joint equilibrium; tensile-coefficient equations
B	8	Three-moment equation; suspension cable	Clapeyron equation; cable parabola, H and length

Symbol & sign quick reference

Symbol	Meaning	Exam note
V	Shear force / base shear depending on context	Define before use.
M	Bending moment	For arches, M = beam moment - $H y$.
H	Horizontal thrust	Appears in arches and cables.
N	Axial/normal force	Truss members carry axial force only under ideal assumptions.
Q	Radial/transverse shear in arch	Resolve section resultant along tangent/normal.
E, I, A	Young modulus, second moment of area, cross-sectional area	EI = flexural rigidity; AE = axial rigidity.
Δ , θ	Deflection, rotation/slope	Unit load: force for Δ ; unit moment for θ .
w	UDL intensity	State whether per horizontal span / projected area as relevant.
L, h, d	Span, arch rise, cable dip	Do not mix rise h and cable dip d.
x	Coordinate / moving unit-load position	Always state origin.

SECTION A

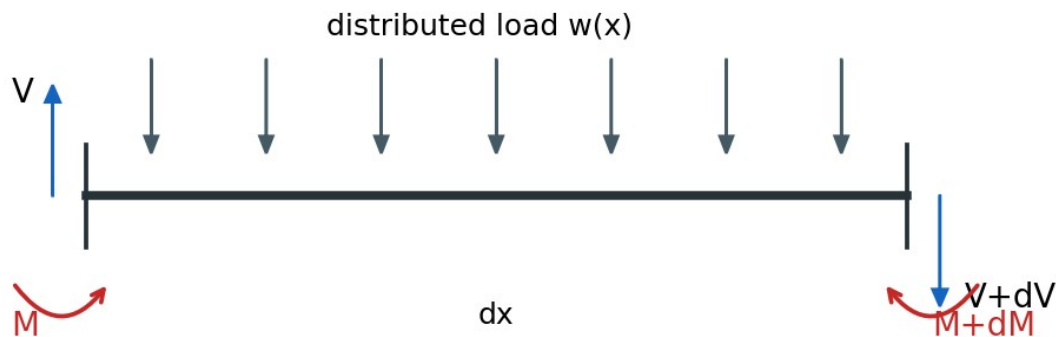
Section A revision logic - Prepare all four sets, then choose the strongest three in the exam. The formula chains below are deliberately arranged from highest-frequency foundations to specialised sub-parts.

SET 1 - Stability, Determinacy, SFD & BMD

Core foundation set **Priority: HIGH**

Master formulas

Topic	Formula / rule	Use / check
2D equilibrium	$\Sigma F_x = 0; \Sigma F_y = 0; \Sigma M = 0$	All determinate reaction and member-force calculations start here.
Support unknowns	Roller = 1, Pin = 2, Fixed = 3	Count independent reaction components.
General determinacy idea	Unknown forces = available independent equilibrium equations $\rightarrow$ determinate	If unknowns exceed equilibrium equations, compatibility is required.
Beam differential relation	$dV/dx = w$ (class-note sign: w positive upward)	Slope of SFD equals load intensity.
Beam differential relation	$dM/dx = V$	Slope of BMD equals shear.
Extremum moment	$V = 0$	An interior maximum/minimum M occurs where shear crosses zero.
Point of contraflexure	$M = 0$ with sign change	BMD crosses the baseline.



Differential beam element used to connect load, shear and bending moment.

Theory answers

1. What is structural analysis?

- It is the systematic evaluation of loads and their effects on a structure.
- The aim is to obtain reactions, internal forces and deformations and to confirm adequate strength, rigidity and safety.
- An idealized structural model is used so that a practical force analysis can be performed.

সহজভাবে: বাস্তব কাঠামোকে idealized model-এ রূপান্তর করে reaction, internal force ও deformation বের করাই structural analysis-এর মূল কাজ।

2. Determinate vs indeterminate; stable vs unstable

Term	Meaning	Key test / consequence
Statically determinate	All required reactions/internal forces are obtainable from equilibrium alone.	No compatibility equation is needed.
Statically indeterminate	Unknown force components exceed the	Need deformation/compatibility relations.

Term	Meaning	Key test / consequence
	independent static equilibrium equations.	
Stable	Support and geometry prevent rigid-body motion/mechanism.	Equilibrium position is maintained.
Unstable	Insufficient/poorly arranged constraints or collapsible geometry permits motion.	Reject even if a simple count looks adequate.

Memory shortcut - Determinate answers "How many equations?"; Stability answers "Can it move as a mechanism?". Count first, inspect geometry second.

3. Free-body diagram (FBD)

- Isolate the entire structure or the required part.
- Replace each support by its reaction component(s).
- Show all external loads, applied couples and internal forces exposed by a cut.
- Write dimensions and choose positive directions before equilibrium equations.

4. Point of inflection vs point of contraflexure

- Point of inflection (elastic curve context): curvature changes sign.
- Point of contraflexure (BMD context): bending moment passes through zero and changes sign.
- In approximate frame analysis, assumed points of inflection are used as zero-moment locations.

Proof / derivation A1 - Load-shear-moment relations

Priority - HIGH - this relation controls shapes of SFD/BMD and is a frequent short derivation.

Take a differential beam element of length dx carrying distributed load $w(x)$. Use the class-note sign convention in which w is positive upward. Let the left-face actions be V and M and the right-face actions be $V+dV$ and $M+dM$.

$$\sum F_y = 0 \rightarrow (V + dV) - V - [\text{sign-consistent load resultant}] = 0$$

After applying the class-note sign convention and dividing by dx :

$$dV/dx = w$$

Now take moments about one face and neglect products of differentials such as $dV \cdot dx$:

$$dM/dx = V$$

Therefore:

- constant $w \rightarrow$ linear V ; zero $w \rightarrow$ constant V ;
- constant $V \rightarrow$ linear M ;
- where $V = 0$, the BMD slope is zero, so M is locally maximum/minimum.

সহজভাবে: লোড ডায়াগ্রামকে একবার integrate করলে SFD, আর SFD-কে integrate করলে BMD-এর পরিবর্তন পাওয়া যায় - এই ধারণাটাই মনে রাখুন।

Common mistakes

Mistake	Correction
Counting reactions but ignoring geometry	A structure can satisfy a count and still be unstable because reactions are parallel/concurrent or the members form a mechanism.
Changing sign convention halfway	Write your positive shear/moment convention once and keep it.
Drawing BMD without checking $dM/dx = V$	Use the shear sign and zero-shear points as a shape check; remember a concentrated couple jumps BMD, not SFD.

Set 1 - 60-second revision grid

Condition	SFD consequence	BMD consequence
$w = 0$	V is constant	M is linear
Constant UDL	V varies linearly	M is parabolic
Concentrated force	V jumps by the force	M remains continuous
Concentrated couple	V is unchanged	M jumps by the couple
$V = 0$ at an interior point	Shear crosses/contacts zero	M is stationary: possible max/min
$M = 0$ with sign change	-	Point of contraflexure

Final Set 1 check - Before leaving a problem: reactions must balance the external loads; every internal hinge has $M=0$; stability requires geometry as well as a correct unknown count; and SFD/BMD shapes must agree with $dV/dx = w$ and $dM/dx = V$.

SET 2 - Plane Truss & Three-Hinged Arch

Axial-force systems and horizontal thrust **Priority: HIGH**

Plane truss - master formulas & rules

Item	Formula / rule	Interpretation
Plane truss determinacy	$b + r = 2j \rightarrow$ determinate (if stable)	b = bars, r = external reactions, j = joints.
Indeterminate count	$b + r > 2j$	Degree by count = $b + r - 2j$, provided stable.
Unstable count	$b + r < 2j$	Insufficient constraints/members.
Joint equilibrium	$\sum F_x = 0; \sum F_y = 0$	Best when joint has at most two unknown member forces.
Section equilibrium	$\sum F_x = 0; \sum F_y = 0; \sum M = 0$	Cut through no more than three unknown member forces.
Member sign	Tension positive; compression negative (if all unknowns initially assumed in tension)	State T/C in final answer.

Rule 1: two non-collinear members
no load/reaction $\rightarrow$ both zero



Rule 2: two collinear + one non-collinear
no load/reaction $\rightarrow$ third member zero

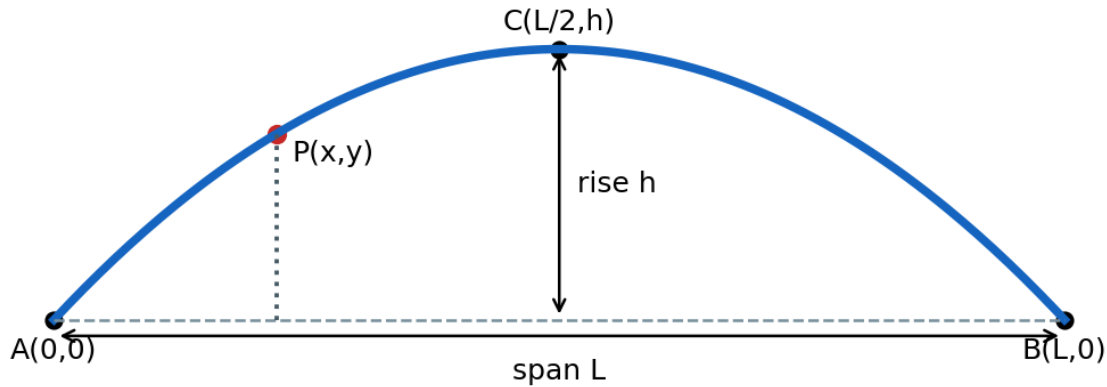


Zero-force member rules by inspection.

Truss assumptions

- Members are connected by smooth/frictionless pins at their ends.
- Loads are applied at joints; member self-weight is neglected or lumped at joints.
- Member centrelines meet at a joint and each member acts as a two-force axial member.
- A force tending to elongate a member is tension; one tending to shorten it is compression.

Three-hinged arch - master formulas



Geometry of a symmetrical three-hinged parabolic arch.

Quantity	Formula	Use
Parabolic ordinate	$y = (4 h x / L^2)(L - x)$	Origin at left springing A; crown C at $(L/2, h)$.
Slope	$dy/dx = (4h/L^2)(L - 2x)$	$\theta = \tan^{-1}(dy/dx)$.
Horizontal thrust	From moment about crown hinge C: $\sum M_C = 0$	Crown hinge moment is zero.
Arch bending moment	$M_{\text{arch}}(x) = M_{\text{beam}}(x) - H y$	Most useful arch shortcut.
Section vertical resultant	$V = R_A - \sum(\text{vertical loads left of section})$	Use in force resolution.
Normal thrust	$N = H \cos\theta + V \sin\theta$	Compression-positive convention shown here.
Radial shear	$Q = V \cos\theta - H \sin\theta$	Sign depends on chosen local axes; sketch axes.
Moment at hinges	$M_A = M_C = M_B = 0$	Three hinges release moment.

Theory: why arch is efficient

- A beam under transverse load develops large bending moment, especially as span increases.
- An arch develops horizontal reaction H; the term $H y$ reduces bending moment.
- A funicular/inverted cable shape can carry the corresponding loading mainly by axial compression.
- Three-hinged arch is statically determinate; two-hinged and fixed arches are statically indeterminate.

Proof / derivation A2 - Equation of a parabolic arch

Priority - HIGH - directly used in many arch numericals.

For a symmetric parabola through $A(0,0)$ and $B(L,0)$, write the ordinate as $y = k x(L-x)$. At the crown $x = L/2$, $y = h$:

$$h = k (L/2)(L/2) = k L^2/4 \rightarrow k = 4h/L^2$$

$$\text{Therefore } y = (4 h x / L^2)(L - x)$$

Differentiate to obtain the tangent slope:

$$dy/dx = (4h/L^2)(L - 2x), \quad \theta = \tan^{-1}(dy/dx)$$

Proof / derivation A3 - Why $M_{\text{arch}} = M_{\text{beam}} - H y$

Cut the arch at any point P(x,y). The vertical reactions and vertical loads create exactly the same bending effect as the corresponding simply supported beam. The horizontal thrust H has lever arm y and produces an opposite moment about P. Hence:

$$M_{\text{arch}} = M_{\text{beam}} - H y$$

সহজভাবে: একই span-এর simply supported beam-এর moment থেকে horizontal thrust-এর moment Hy বাদ দিলেই arch moment পাওয়া যায়।

Proof / derivation A4 - Full-span UDL on a parabolic arch gives zero BM

Priority - HIGH - a classic "show that" proof.

For a symmetrical span L carrying UDL w over the whole horizontal span:

$$R_A = R_B = wL/2$$

At the crown, $M_C = 0$. For the equivalent simply supported beam, crown moment is $wL^2/8$. Therefore:

$$H h = wL^2/8 \rightarrow H = wL^2/(8h)$$

At any section x:

$$M_{\text{beam}}(x) = (w/2) x(L-x)$$

$$H y = [wL^2/(8h)] \cdot [4h x(L-x)/L^2] = (w/2)x(L-x)$$

$$\text{Therefore } M_{\text{arch}}(x) = M_{\text{beam}} - H y = 0 \text{ at every section.}$$

Common mistakes

Mistake	Correction
Using rise h as y at every section	Compute local y from the parabolic equation.
Forgetting horizontal reaction	Three-hinged arch usually has H at both springings.
Resolving N and Q with wrong tangent angle	First compute dy/dx, draw local tangent/normal axes, then resolve.
Truss cut through too many unknowns	Method of sections should normally expose no more than three unknown member forces in 2D.

SET 3 - Wind Load & Earthquake Load

Lateral-load formula chains **Priority: HIGH**

Wind - essential theory

- Wind is natural horizontal motion of air; design wind load depends on wind velocity, building size/shape and terrain exposure.
- The class material distinguishes prevailing, seasonal and local winds.
- Important characteristics: variation of velocity with height, turbulence, statistical probability, vortex shedding and dynamic wind-structure interaction.
- Exposure A: urban/suburban/wooded/hilly with significant obstructions; Exposure B: open terrain with scattered low obstructions; Exposure C: flat unobstructed/coastal/riverside terrain.

Wind formula chain

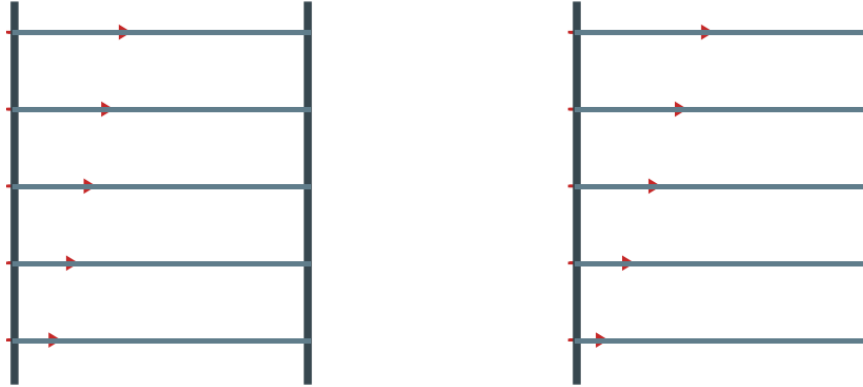
Step	Formula	Meaning
1. Basic speed	V_b	Take the prescribed basic wind speed for the location.
2. Sustained pressure	$q_z = C_c C_l C_z V_b^2$	C_c = velocity-to-pressure conversion coefficient; C_l importance; C_z height/exposure.
3. Design pressure	$p_z = C_G C_p q_z$	C_G = gust response factor; C_p = pressure coefficient.
4. Floor force	$F_i = p_i \times A_{i,tributary}$	Use projected tributary area normal to wind direction.
5. Check	ΣF_i = total applied wind force	Units must remain consistent, usually $\text{kN/m}^2 \times \text{m}^2 = \text{kN}$.

Vortex shedding - Alternating vortices formed behind a bluff body can produce cross-wind oscillation. It matters because the fluctuating transverse action may approach a structural natural frequency.

Earthquake - essential theory

- Magnitude measures energy released by an earthquake; intensity measures shaking at a particular site.
- Base shear = total design lateral earthquake force at the base; storey shear = sum of floor forces at and above a storey; storey drift = relative horizontal displacement between adjacent floors.
- Equivalent Static Force Method starts from a design spectral acceleration and design base shear; Dynamic Response methods include response spectrum and time-history analysis.
- Wind is applied externally on exposed surfaces, whereas earthquake force is primarily an inertia effect associated with structural mass and ground motion.

Wind: pressure x tributary area **Earthquake: base shear distributed by mass & height**



Typical vertical distribution concepts for wind and earthquake floor forces.

Equivalent static force method - formula chain

Quantity	Formula	Notes
Approx. period	$T = C_t h_n^m$	C_t and m depend on structural system.
Damping correction	$\eta = \sqrt{[10/(5+\xi)]} \geq 0.55$	ξ is viscous damping ratio in percent of critical damping.
Normalized spectrum, $0 \leq T \leq T_B$	$C_s = S[1 + (T/T_B)(2.5\eta - 1)]$	Site-dependent S , T_B , T_C , T_D .
$T_B \leq T \leq T_C$	$C_s = 2.5 S \eta$	Constant spectral acceleration region.
$T_C \leq T \leq T_D$	$C_s = 2.5 S \eta (T_C/T)$	Descending branch.
$T_D \leq T \leq 4 \text{ s}$	$C_s = 2.5 S \eta (T_C T_D/T^2)$	Long-period branch.
Design spectral acceleration	$S_a = (2/3)(Z/I/R) C_s$	Apply the code lower-bound requirement where specified.
Base shear	$V = S_a W$	W = total seismic weight.
Floor force	$F_x = V [w_x h_x^k / \sum(w_i h_i^k)]$	$k = 1$ for $T \leq 0.5 \text{ s}$; $k = 2$ for $T \geq 2.5 \text{ s}$; interpolate between.
Storey shear	$V_x = \sum F_i$ for $i = x \dots n$	Sum floor forces at and above the storey.

Bangladesh seismic zoning values shown in the lecture material

Zone	Z	Lecture description
1	0.12	Low
2	0.20	Moderate
3	0.28	Severe
4	0.36	Very severe

Derivation / logical check A5 - Floor-force distribution closes to base shear

Because each floor force uses the same normalizing denominator:

$$F_x = V \cdot [w_x h_x^k / \sum(w_i h_i^k)]$$

Summing over all floors:

$$\Sigma F_x = V \cdot [\Sigma(w_x h_x^k) / \Sigma(w_i h_i^k)] = V$$

Exam check - After calculating the floor forces, always add them. Their total must equal the design base shear V (within rounding).

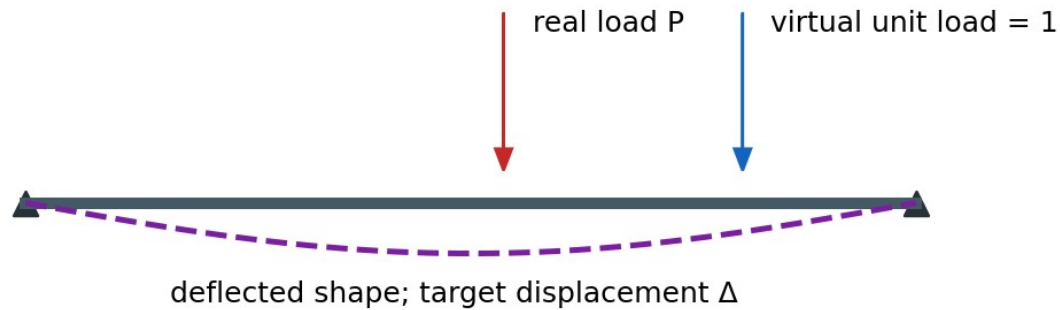
Common mistakes

Mistake	Correction
Using gross floor area instead of wind projected tributary area	For wind, use area normal to the wind direction and the correct tributary height/width.
Mixing C_z and C_G tables	C_z belongs to sustained pressure; C_G enters design pressure.
For earthquake, distributing V equally	Use $w_i h_i^k$ weighting; higher/heavier floors generally attract more lateral force.
Using total dead load without the problem's seismic weight instruction	Use the seismic weight specified in the question/data.

SET 4 - Deflection: Unit Load / Virtual Work & Conjugate Beam

Displacement methods **Priority: VERY HIGH**

Virtual-work master formulas



Real loading produces deformation; a virtual unit action extracts the desired displacement component.

Application	Formula	How to create the virtual system
Beam/frame deflection	$\Delta = \int (M m / EI) dx$	Apply a unit force at the target point in the target direction.
Beam/frame rotation	$\theta = \int (M m_{\theta} / EI) dx$	Apply a unit couple moment at the target point in the target sense.
Truss displacement	$\Delta = \sum (N n L / AE)$	Unit force at the target joint; N = real axial force, n = virtual axial force.
Truss temperature effect	$\Delta_T = \sum n (\alpha \Delta T L)$	Use virtual member forces n and free thermal elongation $\alpha \Delta T L$.
Fabrication error	$\Delta_f = \sum n (\delta L)$	δL = imposed member length error.

Theory: principle of virtual work

- For a deformable structure in equilibrium, virtual external work equals virtual internal work for compatible small deformations.
- The virtual load is imaginary and usually has unit magnitude; it is used only to select the desired displacement/rotation.
- The method applies to beams, frames and trusses and can include load, temperature change or fabrication error effects.

Proof / derivation A6 - Unit-load formula for beams and frames

Priority - VERY HIGH - directly appears as a proof question and supports many numericals.

Let the real loads create bending moment $M(x)$. To find displacement Δ at point A, apply a virtual unit load in the direction of Δ . The virtual system creates moment $m(x)$. For an element dx :

$$\text{Real rotation of the element: } d\theta = [M/(EI)] dx$$

Virtual internal work of m through this real rotation is:

$$dW_{int} = m d\theta = [M m/(EI)] dx$$

The external virtual work of the unit force through the real displacement is:

$$W_{ext} = 1 \cdot \Delta$$

Equating external and internal virtual work and integrating over all members:

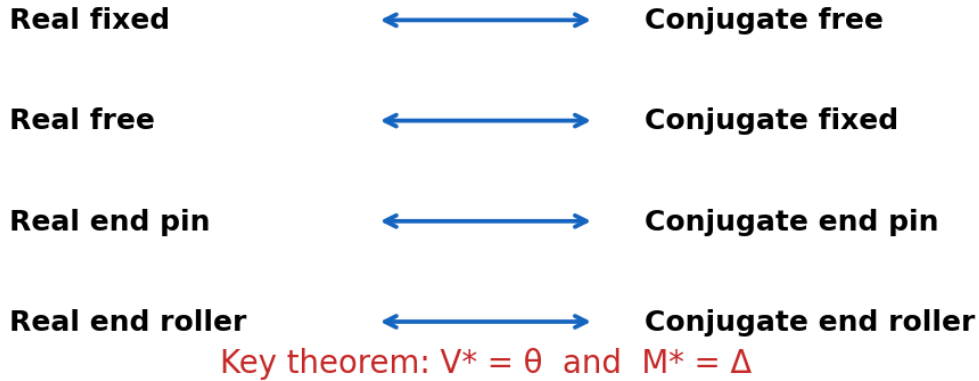
$$\Delta = \int M m/(EI) dx$$

For a truss member, real axial deformation is $\delta L = NL/(AE)$; therefore:

$$\Delta = \sum n \cdot NL/(AE) = \sum N n L/(AE)$$

Conjugate beam - master theory & formulas

Support transformation memory map



- A conjugate beam is a fictitious beam of the same length as the real beam.
- It is loaded by the real beam's M/EI diagram.
- Shear in the conjugate beam equals slope in the real beam; moment in the conjugate beam equals deflection in the real beam.
- With the lecture sign convention, positive conjugate shear denotes positive (counterclockwise) slope and positive conjugate moment denotes upward deflection.

$$V^* = \theta, \quad M^* = \Delta, \quad w^* = M/EI$$

Real beam condition	Conjugate beam condition	Reason
Fixed end	Free end	$\theta = 0$ and $\Delta = 0$ in real beam $\rightarrow V^*$ and M^* need not be restrained there.
Free end	Fixed end	Real θ and Δ are free/nonzero $\rightarrow$ conjugate must develop V^* and M^* reactions.
End pin	End pin	Real $\Delta = 0$ but θ may exist $\rightarrow M^* = 0$, V^* may exist.
End roller	End roller	Same endpoint condition as pin for vertical-bending problem.
Internal hinge / simple support relation	Use the corresponding internal support transformation shown in class slide	Track whether slope is continuous/discontinuous and whether deflection is zero.

Proof / derivation A7 - Conjugate beam theorem

Priority - HIGH - compact proof by differential analogy.

For the real beam elastic curve:

$$d\theta/dx = M/EI, \quad dy/dx = \theta$$

For an ordinary beam under load w^* :

$$dV^*/dx = w^*, \quad dM^*/dx = V^*$$

Choose the conjugate loading $w^* = M/EI$. Then the differential chains are identical:

$$dV^*/dx = d\theta/dx \rightarrow V^* = \theta \text{ (with matching boundary conditions)}$$

$$dM^*/dx = V^* = \theta = dy/dx \rightarrow M^* = y = \Delta$$

সহজভাবে: Real beam-এর M/EI diagram-কে conjugate beam-এর load থরলে, conjugate shear = real slope এবং conjugate moment = real deflection।

Common mistakes

Mistake	Correction
Using the real load as conjugate load	Conjugate beam load is M/EI , not the original loading.
Applying unit force for a rotation	Use a unit moment/couple to extract rotation.
Ignoring different EI segments	Integrate or construct M/EI piecewise.
Using wrong support transformation	Write the real boundary conditions (θ and Δ) first, then map them.

SECTION B

Section B revision logic - The later-analysis set is formula-dense. Learn the assumptions first; a correct approximate-analysis or ILD solution is impossible if the assumed release/inflection locations are wrong.

SET 5 - Influence Line Diagram (ILD)

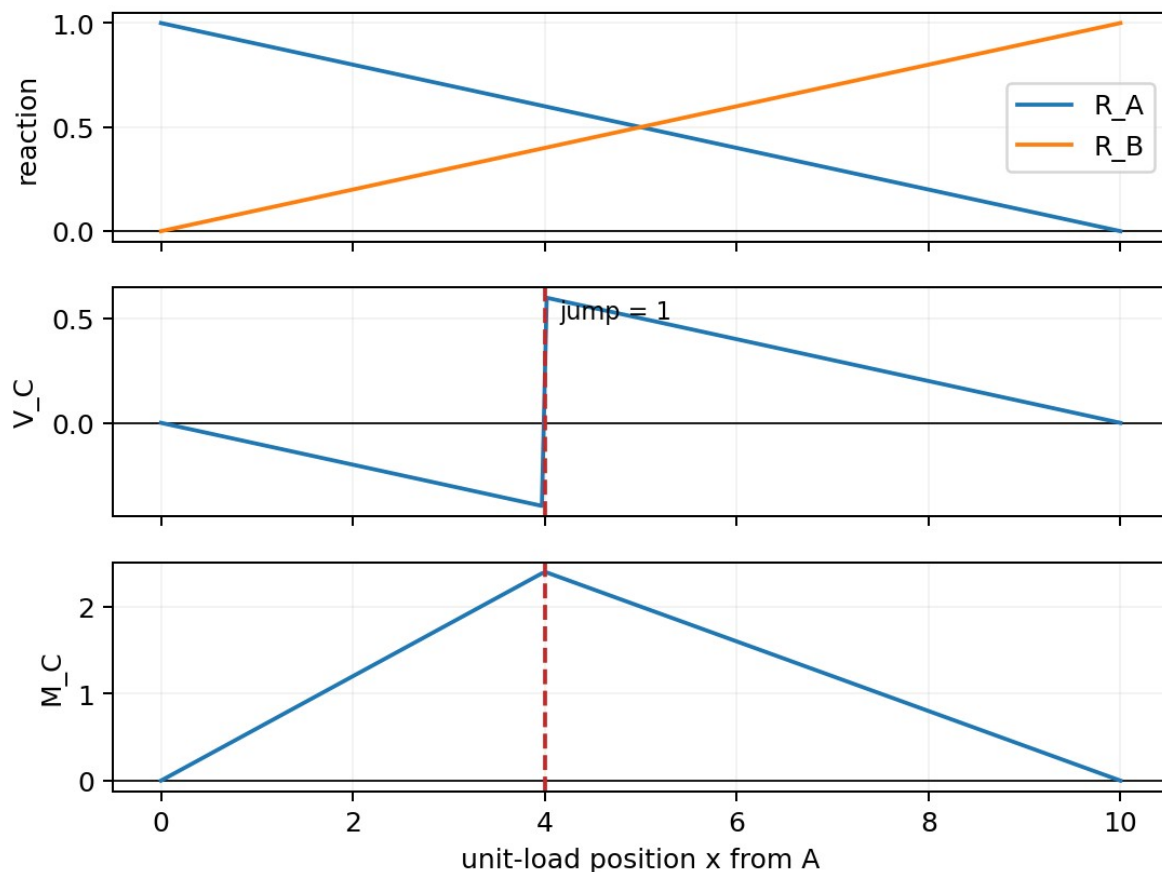
Moving-unit-load response functions **Priority: VERY HIGH**

Core definition & methods

- An influence line is the graph of a response function of a structure as a downward unit load moves across the structure.
- Typical response functions: support reaction, shear, bending moment and sometimes axial force.
- The class note uses two approaches: (i) equilibrium method and (ii) Muller-Breslau principle.

Simply supported beam ILD formulas

Basic ILD ordinates for a simply supported beam



Analytical ILDs for a simple beam; shear ordinate has a unit jump at the section.

For a simply supported span L with a unit load at x from A:

$$R_A(x) = (L - x)/L, \quad R_B(x) = x/L$$

At section C located a from A:

$$\begin{aligned} V_C &= -x/L & \text{for } 0 \leq x < a \\ V_C &= (L - x)/L & \text{for } a < x \leq L \\ M_C &= x(L-a)/L & \text{for } 0 \leq x \leq a \\ M_C &= a(L-x)/L & \text{for } a \leq x \leq L \end{aligned}$$

Shear jump rule - The ILD for shear at a section has a jump of exactly 1 at that section because the moving load itself crosses the cut.

Derivation B1 - Reaction, shear and moment ILDs by equilibrium

Priority - HIGH - this is the safest method for complicated determinate beams with hinges/overhangs.

Place a downward unit load at a general coordinate x . Whole-beam moment equilibrium about A gives:

$$R_B L - 1 \cdot x = 0 \rightarrow R_B = x/L$$

$$R_A = 1 - R_B = (L-x)/L$$

For shear at C, use a cut at C. If the unit load is left of C, the left free body contains the unit load; if it is right of C, it does not. This produces the two expressions and the unit jump shown above.

For moment at C, take moments of the left free body about C. The result is piecewise linear, with peak ordinate:

$$M_{C, \text{max ordinate at } x=a} = a(L-a)/L$$

Theory / proof B2 - Muller-Breslau principle

Priority - HIGH - common theory question and powerful sketching principle.

Class-note statement in practical form: the influence line for a force or moment response is proportional to the deflected shape of the structure obtained after removing the restraint corresponding to that response and imposing a unit displacement or unit rotation in the positive direction of that response.

Desired ILD	Release / unit action
Reaction at a support	Remove the support restraint in that direction and impose unit displacement.
Shear at a section	Cut/release the shear transfer and impose unit relative transverse displacement.
Moment at a section	Insert a hinge/release moment and impose unit relative rotation.
Reaction moment at a fixed support	Release the rotational restraint and impose unit rotation.

সহজভাবে: যে response-এর ILD চাই, সেই restraint-টা release করে positive direction-এ unit displacement/rotation দিন; released shape-টাই ILD-এর shape বলে দেবে।

ILD workflow for beams/frames with internal hinges

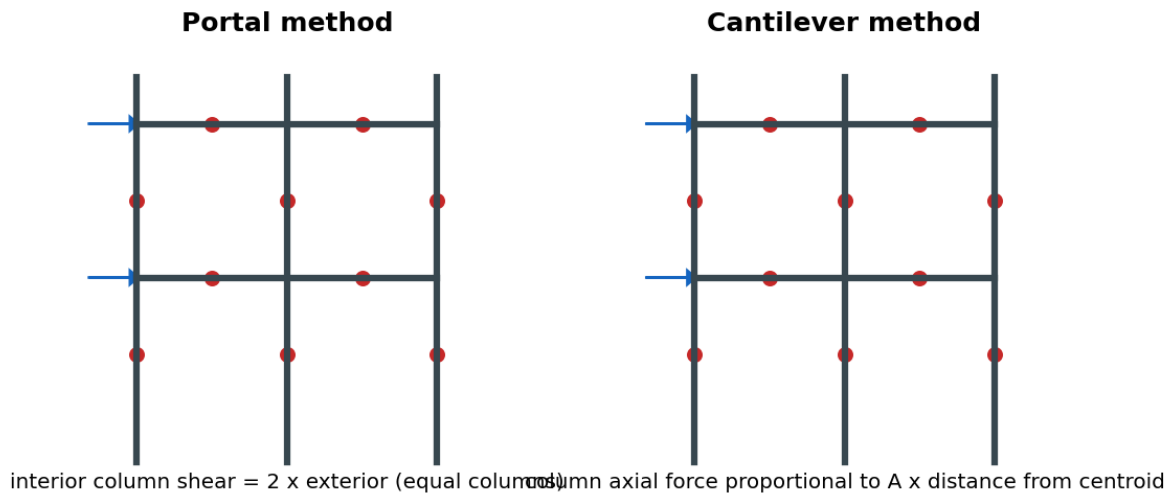
- Mark every support, internal hinge and response location.
- Move the unit load region by region; internal hinges often change the governing equilibrium equations.
- Derive ordinates symbolically in x for each region.
- Evaluate ordinates at all key points and join linearly where the structure is statically determinate and the expressions are linear.
- For shear, plot separate just-left and just-right ordinates at the section.

Common mistakes

Mistake	Correction
Treating ILD as SFD/BMD under a fixed load	ILD ordinate is a response versus moving-load position, not position along the beam under one fixed loading.
Forgetting internal hinge condition	Moment at an internal hinge is zero; use it to generate an extra equilibrium relation.
Missing shear jump	Plot left and right ordinates separately; jump magnitude is 1 for a unit load.
Joining nonadjacent key ordinates without checking regions	Derive piecewise equations first.

SET 6 - Approximate Analysis of Building Frames

Portal, cantilever, factor and vertical-load methods **Priority: VERY HIGH**



Assumed inflection locations and the key force-distribution concept.

Why approximate analysis works

- Indeterminate building frames have more unknown actions than static equilibrium equations.
- Approximate methods reduce the degree of indeterminacy by assuming locations of zero moment and/or force-distribution patterns.
- The assumptions must be stated before calculations; they are part of the answer.

Portal method - assumptions & formulas

Item	Portal-method statement
Inflection points	At the centre of each girder and each column.
Storey shear distribution	For a regular equal-column frame, each interior column carries twice the shear of an exterior column.
Exterior-column shortcut	If there are m columns in the storey: $V_{ext} = V_{storey} / [2(m-1)]$.
Interior-column shortcut	$V_{int} = 2 V_{ext} = V_{storey} / (m-1)$.
Column end moment	$M_{end} = V_{column} \times (\text{storey height}/2)$ when inflection is at mid-height.
Beam end moment	Obtain from joint moment equilibrium; beam inflection at midspan fixes opposite-end relationship.

Cantilever method - assumptions

- A point of inflection occurs at the centre of each beam and column.
- At any storey, axial stress in each column is proportional to the horizontal distance of that column from the centroidal axis of all columns of that storey.
- For unequal column areas, the force distribution must include A_i .

Derivation B3 - Cantilever-method column axial forces

Priority - HIGH - turns the qualitative cantilever assumption into a direct formula.

Let the centroidal axis of the column group be:

$$\bar{x} = \Sigma(A_i x_i) / \Sigma A_i$$

Let $r_i = x_i - \bar{x}$. The cantilever assumption gives axial stress $\sigma_i \propto r_i$, hence axial force:

$$F_i = C A_i r_i$$

The column axial forces resist the storey overturning moment M_s :

$$M_s = \Sigma(F_i r_i) = C \Sigma(A_i r_i^2)$$

$$\text{Therefore } C = M_s / \Sigma(A_i r_i^2)$$

$$F_i = M_s A_i r_i / \Sigma(A_j r_j^2)$$

Interpretation - Columns farthest from the centroidal axis attract the largest tension/compression; a larger area also attracts a larger force.

Approximate vertical-load analysis - class-note assumptions

Assumption / quantity	Class-note rule
Beam axial force	Neglected (taken as zero).
Beam inflection point	At one-tenth of the girder/beam span measured from each support.
Joint unbalanced moment	Distributed among columns meeting the joint according to their relative stiffness.
Column bending stiffness	$K_b = EI/L$.
Column axial stiffness	$K_a = AE/L$.
Workflow	Find girder end moments -> transfer/distribute to columns -> obtain column shear from end moments -> draw SFD/BMD.

Factor method - safe formula summary from the class sheet

- Compute member stiffness parameter $K = I/L$ for each girder and column.
- At each joint, obtain the girder/column factors from the relative K values of members meeting at that joint.
- The class sheet uses a complementary column factor $c = 1 - g$; at a fixed base the column factor is taken as 1.0.
- Use the tabulated joint/storey factors to distribute moments, then obtain column shear and complete SFD/BMD by equilibrium.

No-error note - The exact factor-table arithmetic depends on the class-sheet layout and member stiffness pattern. In an exam, reproduce the same K , g and c table format used in class before multiplying by storey/joint moments.

Portal vs cantilever - theory comparison

Aspect	Portal method	Cantilever method
Best suited conceptually	Low/medium-rise regular frames dominated by shear-type behaviour.	Taller slender frames dominated by overturning/cantilever-type behaviour.
Key distribution assumption	Interior column shear = $2 \times$ exterior (regular equal columns).	Column axial stress $\propto$ distance from centroidal axis.
POI	Midspan of beams and mid-height of columns.	Midspan of beams and mid-height of columns.
Main calculated first	Column shears.	Column axial forces.

Common mistakes

Mistake	Correction
Starting equilibrium before writing assumptions	State method assumptions first; they create the extra equations.
Portal: dividing storey shear equally among all columns	Interior columns take twice exterior shear for the standard regular-frame assumption.
Cantilever: using geometric centroid for unequal areas	Use area-weighted centroid $\bar{x} = \Sigma Ax / \Sigma A$.
Forgetting that story overturning moment changes by level	Compute the correct moment of all lateral loads above/belonging to the cut.

SET 7 - Mill Bent / Bridge Portal & Space Truss

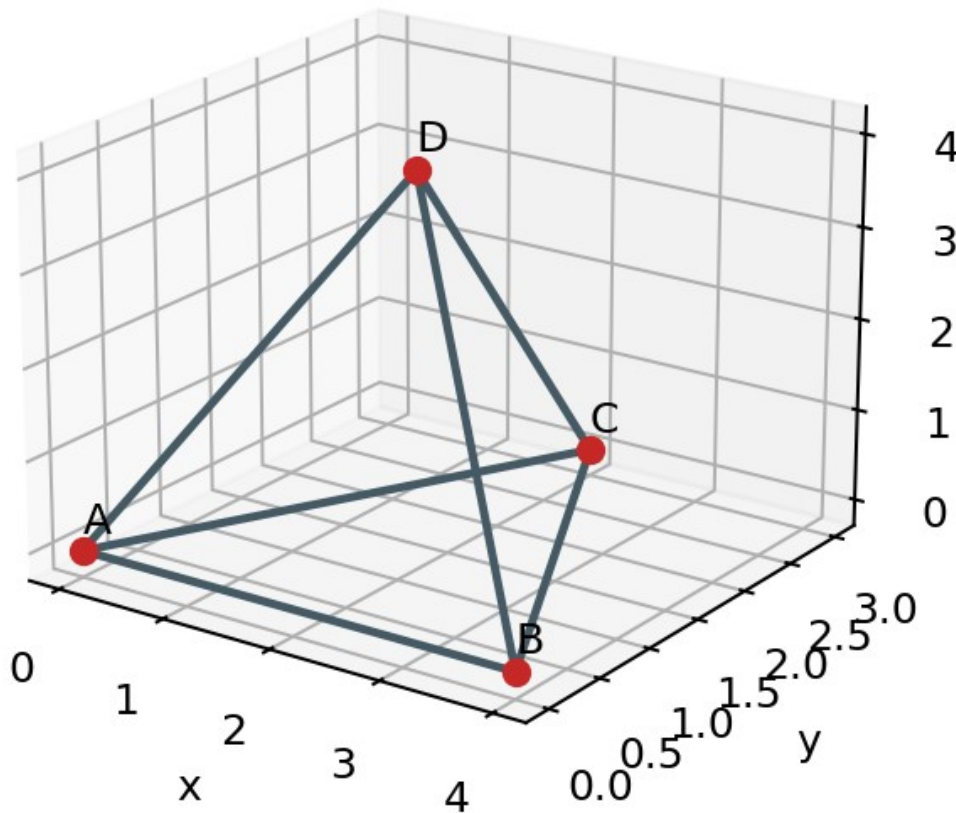
Portal-type industrial frames and 3D trusses **Priority: HIGH**

Mill bent / bridge portal - class-note assumptions

- Horizontal support reactions are assumed equal for the simple bridge-portal/mill-bent approximation shown in class.
- For fixed-base columns with an unsupported height, a point of inflection is assumed at the end of the unsupported height.
- After support reactions are obtained, leg shears and end moments are found from equilibrium and the SFD/BMD of the legs is drawn.
- A mill bent is treated as a portal-type frame that may include a trussed roof/girder; analyse the lateral-load path through the frame legs and roof system.

Space truss - theory and equations

Space truss: 3D axial-force members



A space truss requires three force-equilibrium equations at a joint and six equilibrium equations for the whole 3D body.

- Members of a three-dimensional truss are connected by universal/pin-type joints so that ideal members carry axial force only.
- Whole-structure equilibrium in 3D provides six scalar equations.
- Class notes list methods including projection of forces on axes, projection of joints on a plane, tensile-coefficient method and moment about an axis.

$$\Sigma F_x = 0, \Sigma F_y = 0, \Sigma F_z = 0, \Sigma M_x = 0, \Sigma M_y = 0, \Sigma M_z = 0$$

Direction-cosine form

For member ij with coordinate differences $\Delta x, \Delta y, \Delta z$:

$$L_{ij} = \sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}$$

$$\lambda = \Delta x/L, \quad \mu = \Delta y/L, \quad \nu = \Delta z/L$$

$$\vec{F}_{ij} = F_{ij}(\lambda \mathbf{i} + \mu \mathbf{j} + \nu \mathbf{k})$$

Derivation B4 - Tensile-coefficient method

Priority - HIGH - this is the compact coordinate method seen in the class note.

Define the tensile coefficient of member ij as:

$$t_{ij} = F_{ij} / L_{ij} \rightarrow F_{ij} = t_{ij} L_{ij}$$

The x-component of the member force acting at joint i is:

$$F_x = F_{ij} (x_j - x_i) / L_{ij} = t_{ij} (x_j - x_i)$$

Similarly for y and z. Thus joint i equilibrium becomes:

$$\sum t_{ij} (x_j - x_i) + X_i = 0$$

$$\sum t_{ij} (y_j - y_i) + Y_i = 0$$

$$\sum t_{ij} (z_j - z_i) + Z_i = 0$$

Solve the three joint equations for the tensile coefficients, then multiply each by its member length. A positive coefficient under a tension-positive convention means tension; negative means compression.

Space-truss stability/determinacy note

- Do not apply the 2D relation $b+r=2j$ to a 3D truss.
- Use the three translational joint-equilibrium equations and inspect geometry for mechanisms; the class questions generally proceed by reactions plus joint/coordinate equilibrium.

Common mistakes

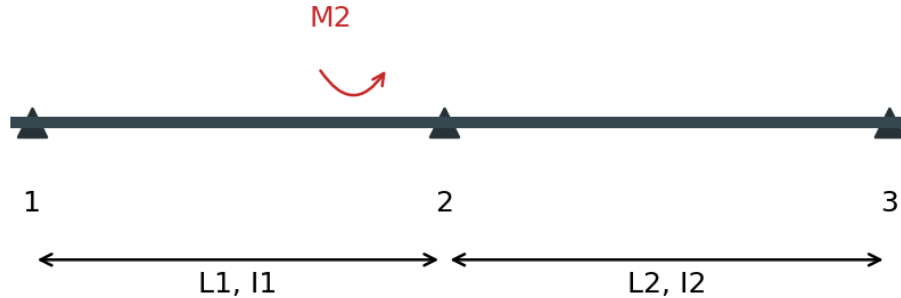
Mistake	Correction
Mixing plan and elevation coordinates	Assign a single x-y-z coordinate to every joint before equations.
Using 2D member length	Use the full 3D distance $\sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}$.
Dropping the sign of tensile coefficient	The coefficient sign carries T/C information.
Mill bent: treating roof truss and legs as unrelated	Follow the lateral force transfer through the complete bent and use joint/leg equilibrium.

SET 8 - Three-Moment Equation & Suspension Cable

Continuous-beam compatibility and funicular cable action **Priority: VERY HIGH**

Three-moment equation - master formula

Clapeyron three-moment relation couples M_1 , M_2 , M_3



Three consecutive supports and two spans used in the three-moment relation.

For consecutive supports 1-2-3, spans L_1 and L_2 , flexural rigidities EI_1 and EI_2 , point loads P with k = distance from the left support/span, full-span UDLs w_1 and w_2 , and settlements Δ_1 , Δ_2 , Δ_3 , the class-note form is:

$$\begin{aligned} M_1 L_1/I_1 + 2M_2(L_1/I_1 + L_2/I_2) + M_3 L_2/I_2 = \\ - \Sigma[P_1 L_1 k_1(1-k_1^2)/I_1] - \Sigma[P_2 L_2 k_2(1-k_2^2)/I_2] \\ - w_1 L_1^3/(4I_1) - w_2 L_2^3/(4I_2) - 6E[(\Delta_1-\Delta_2)/L_1 + (\Delta_3-\Delta_2)/L_2] \end{aligned}$$

Definition of k - For a point load in a span, k = distance of that point load from the LEFT support of that span / span length.

Useful special cases

Condition	Simplification
Simple end support	End support moment = 0.
Internal hinge	Moment at hinge = 0.
No support settlement	All Δ terms are zero.
Constant I	Cancel the common I after writing the equation consistently.
No load in a span	All point-load and UDL terms for that span are zero.
After support moments	Treat each span as a beam with known end moments; use equilibrium for reactions, then draw SFD/BMD.

Derivation B5 - Three-moment equation (class-note compatibility route)

Priority - VERY HIGH - the main formal derivation in Section B.

Consider two adjacent spans meeting at middle support 2. Rotation continuity at support 2 requires the rotations contributed by loading, end moments and support settlement from the two spans to be compatible. The class note writes the end-moment rotation contributions in the familiar form:

$$\begin{aligned} \text{rotation from own-end moment } M \text{ over span } L: & \quad M L/(3EI) \\ \text{rotation from far-end moment } M_{\text{far}} \text{ over span } L: & \quad M_{\text{far}} L/(6EI) \end{aligned}$$

For a point load P located at kL from the left support, the load-induced rotation term used in the class-note derivation leads, after multiplication by $6E$, to:

$$P L k(1-k^2)/I$$

For a full-span UDL, the corresponding term after multiplication by $6E$ is:

$$w L^3/(4I)$$

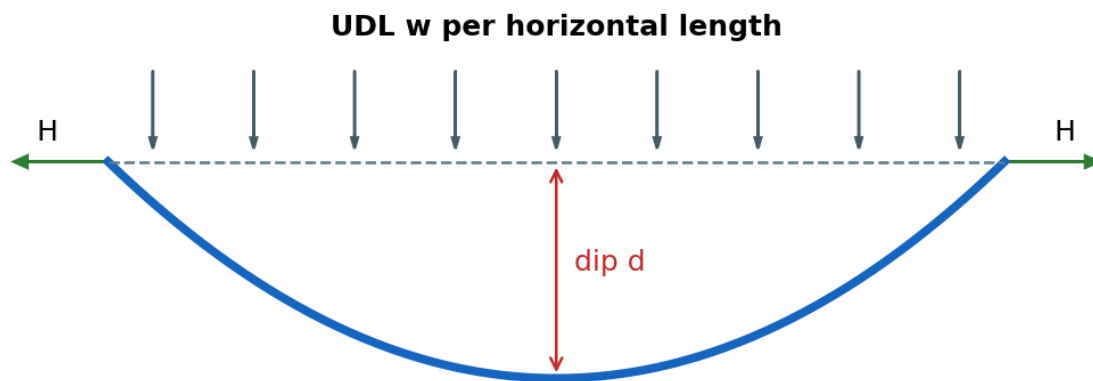
Support settlement contributes a chord-rotation term. For the left and right spans adjacent to support 2:

$$(\Delta_1 - \Delta_2)/L_1 \quad \text{and} \quad (\Delta_3 - \Delta_2)/L_2$$

Add the rotations at support 2, enforce compatibility, substitute the end-moment/load/settlement terms, and multiply through by $6E$. Collecting M_1 , M_2 and M_3 gives the three-moment equation printed above.

সহজভাবে: Derivation-এর মূল কথা: middle support-এ slope continuity + end moment-এর rotation + load-এর rotation + settlement chord rotation = compatibility condition ।

Suspension cable - theory



A perfectly flexible cable under UDL over the horizontal span takes a parabolic profile.

- A cable is flexible and carries load essentially by direct tension; it cannot resist bending moment as a beam does.
- Under concentrated loads the cable profile is a funicular polygon.
- Under UDL per horizontal length, a cable between level supports is parabolic.
- Maximum tension generally occurs at a support because both horizontal and vertical tension components are largest there.
- Suspension bridges use main cables, hangers, towers, deck/stiffening system and anchorage/back-stays; class notes list long span and good earthquake performance as advantages, with low deck stiffness/vibration sensitivity as disadvantages.

Cable under concentrated loads - same support level

Quantity	Formula / equation
Horizontal thrust	$H_A = H_B = H$
Vertical reactions	$R_A + R_B = \Sigma W$
Support moment equation	Take moments of all external vertical loads to obtain R_A , R_B .
Zero BM at any cable point i	$R_A x_i - \Sigma [W_j(x_i - x_j)] - H y_i = 0$ (include only loads left of point i)
Support tension	$T_A = \sqrt{(R_A^2 + H^2)}$, $T_B = \sqrt{(R_B^2 + H^2)}$
Cable length for polygonal segments	$L_{\text{cable}} = \Sigma \sqrt{(\Delta x)^2 + (\Delta y)^2}$

Cable under UDL - same support level

Quantity	Formula
Vertical reactions	$R_A = R_B = wL/2$
Horizontal thrust	$H = wL^2/(8d)$
Cable ordinate (dip downward)	$y = 4d x(L-x)/L^2$
Maximum tension	$T_{\max} = \sqrt{[(wL/2)^2 + H^2]}$
Equivalent closed form	$T_{\max} = (wL/8d) \sqrt{L^2 + 16d^2}$
Approximate cable length	$L_c \approx L + 8d^2/(3L)$

Derivation B6 - Parabolic cable and horizontal thrust under UDL

Priority - VERY HIGH - repeated formula and proof family.

For equal-level supports and UDL w per horizontal length, symmetry gives:

$$R_A = R_B = wL/2$$

At the lowest point (centre) the bending moment of a perfectly flexible cable is zero. Taking moments about the centre:

$$(wL/2)(L/2) - (wL/2)(L/4) - Hd = 0$$

$$H = wL^2/(8d)$$

At a general point x from A with dip y :

$$R_A x - (wx)(x/2) - Hy = 0$$

$$y = [wx(L-x)]/(2H) = 4d x(L-x)/L^2$$

Thus the cable shape is parabolic.

Derivation B7 - Approximate length of a parabolic cable

For the parabola $y = 4d x(L-x)/L^2$:

$$dy/dx = (4d/L^2)(L - 2x)$$

Arc length is:

$$L_c = 2 \int_0^{L/2} \sqrt{1 + (dy/dx)^2} dx$$

For ordinary sag where $(dy/dx)^2$ is not large, use $\sqrt{1+u} \approx 1 + u/2$ and integrate:

$$L_c \approx L + 8d^2/(3L)$$

Different support levels - solution equations

For UDL w over horizontal span L , if supports are at different levels, use the same equilibrium principle but do not assume equal vertical reactions. Let support level difference be h and let the lowest point be a from A with dip d measured from the A-level reference used in the sketch.

$$H_A = H_B = H, \quad R_A + R_B = wL$$

Whole-cable moment equilibrium supplies the third support relation (include the $H \cdot h$ moment with the correct sign from the sketch).

$$\text{At the lowest point: } R_A a - Hd - wa^2/2 = 0$$

Safest exam method - For different-level supports, draw the geometry and derive the sign of $H \cdot h$ from your own moment equation instead of memorising a sign. Then locate the lowest point from zero vertical tension component and compute $T = \sqrt{(H^2 + V^2)}$.

Extra theory - cable under self-weight

- The class reference notes that a cable carrying its own weight takes a catenary, not a parabola.
- With H the horizontal component and w the self-weight per unit cable length, the differential relation shown is $w\sqrt{1+(dy/dx)^2} = H \frac{d^2y}{dx^2}$.

Common mistakes

Mistake	Correction
Three-moment: measuring k from the wrong support	k is measured from the left support of that span as used in the class note.
Forgetting support settlement sign	Write the actual Δ values with a clear positive direction before substituting.
Cable: treating H as varying along the cable	With only vertical loading, the horizontal tension component H is constant.
Cable: using beam BM as nonzero in the cable	A perfectly flexible cable cannot carry bending; use $M = 0$ to obtain H /ordinates.
Using parabolic length formula for a concentrated-load polygon	For concentrated loads, sum the straight segment lengths exactly.

APPENDIX A - Ultra-Compact Formula Sheet

Use - This is the last 10-minute revision page. The detailed derivations are in the sets above.

Set	Must-remember formulas
A1	$\Sigma F_x=0$, $\Sigma F_y=0$, $\Sigma M=0$; $dV/dx=w$; $dM/dx=V$; $V=0$ at extremum M .
A2	Truss: $b+r$ vs $2j$; Arch: $y=4hx(L-x)/L^2$; $M=M_{\text{beam}}-Hy$; $H=wL^2/(8h)$ for full-span UDL parabola.
A3	Wind: $q_z=C_c C_i C_z V_b^2$; $p_z=C_G C_p q_z$; EQ: $T=C_t h^m$; $S_a=(2/3)(Z/I/R)C_s$; $V=S_a W$; $F_x=V w_{xh} x^k / \Sigma w_{ih} i^k$.
A4	$\Delta_{\text{beam}}=JMm/EI dx$; $\Delta_{\text{truss}}=\Sigma NnL/AE$; conjugate: $w^*=M/EI$, $V^*=\theta$, $M^*=\Delta$.
B5	Simple beam: $R_A=(L-x)/L$, $R_B=x/L$; shear ILD jumps 1; $M_C^{\text{peak}}=a(L-a)/L$.
B6	Portal: midspan/midheight POI, interior shear=2 exterior; Cantilever: $\bar{x}=\Sigma Ax/\Sigma A$, $F_i=M_s A_i r_i/\Sigma A r_i^2$; vertical-load $K_b=EI/L$, $K_a=AE/L$.
B7	Space truss: 6 global equilibrium equations; $L=\sqrt{(\Delta x^2+\Delta y^2+\Delta z^2)}$; $t=F/L$; joint equations $\Sigma t\Delta x+X=0$ etc.
B8	Three-moment general equation; Cable UDL: $H=wL^2/(8d)$, $y=4dx(L-x)/L^2$, $T_{\text{max}}=\sqrt{[(wL/2)^2+H^2]}$, $L_c \approx L+8d^2/(3L)$.

Quick load-shape rules

Condition	SFD / BMD implication
No distributed load	SFD horizontal/constant.
Constant UDL	SFD linear; BMD parabolic.
Concentrated force	SFD jumps by the force; BMD remains continuous.
Concentrated couple	BMD jumps by the couple; SFD unchanged.
$V=0$	BMD has a stationary/max-min point.
M changes sign through zero	Point of contraflexure.

APPENDIX B - Theory Question Checklist

Set	Theory to be able to write without notes
A1	Define structure and structural analysis; idealization; FBD; static equilibrium; determinate/indeterminate; stable/unstable; POI/POC.
A2	Define truss; assumptions; method of joints vs sections; zero-force members; arch definition/types; advantages/functions; beam vs arch.
A3	Wind types/characteristics; sustained vs design pressure; exposure A/B/C; vortex shedding; earthquake magnitude vs intensity; base/storey shear and drift; static vs dynamic seismic methods; wind vs earthquake.
A4	Deflection causes; virtual-work principle; unit load meaning; define

Set	Theory to be able to write without notes
	conjugate beam; real-conjugate support relation; sign convention.
B5	Define ILD; response functions; equilibrium method; state/explain Muller-Breslau principle; shear jump.
B6	Why approximate analysis; portal assumptions; cantilever assumptions; vertical-load assumptions; portal vs cantilever; factor-method stiffness concept.
B7	Mill bent/bridge portal assumptions; define space truss; 3D equilibrium; solution methods; tensile coefficient.
B8	State three-moment equation; define k and settlement terms; cable characteristics; suspension bridge components; advantages/disadvantages; same/different level cable; parabolic vs catenary shape.

APPENDIX C - Proof / Derivation Checklist

Priority	Derivation / proof	Final result to reach
VERY HIGH	Unit-load method for beam/frame	$\Delta = \int Mm/EI \, dx$
VERY HIGH	Three-moment equation	General Clapeyron relation including loads and settlements
VERY HIGH	UDL cable parabola and horizontal thrust	$H = wL^2/(8d)$; $y = 4dx(L-x)/L^2$
HIGH	Load-shear-moment relations	$dV/dx = w$; $dM/dx = V$
HIGH	Parabolic arch equation	$y = 4hx(L-x)/L^2$
HIGH	Parabolic arch under full UDL	$M_{\text{arch}} = 0$ at every section
HIGH	Conjugate beam theorem	$V^* = \theta$; $M^* = \Delta$
HIGH	Simple-beam ILD by equilibrium	R_A, R_B, V_C, M_C piecewise
HIGH	Muller-Breslau principle	Released-structure deflected shape gives ILD shape
HIGH	Cantilever-method axial-force distribution	$F_i = M_s A_i r_i / \sum A r^2$
HIGH	Space-truss tensile coefficient	$t=F/L$ and $\sum t\Delta x + X = 0$ etc.
MEDIUM	Parabolic cable length approximation	$L_c \approx L + 8d^2/(3L)$
MEDIUM	Earthquake floor-force sum check	$\sum F_i = V$

APPENDIX D - Examiner-Friendly Presentation Checklist

- Write the method/assumption before equations: "Using portal method...", "Using Muller-Breslau...", "Using unit-load method..."
- Draw FBDs and local axes. For arch N/Q and space truss x-y-z problems, the sketch prevents sign errors.
- Define every non-obvious symbol once, especially k in the three-moment equation and d/h in cable/arch formulas.
- Keep units beside intermediate results. Pressure $\times$ area must become force; EI must use compatible length units in deflection.
- At the end, perform one physical check: reactions balance loads, $\sum$ floor seismic force = base shear, hinge moment = 0, cable H constant, conjugate support condition satisfied.
- If a sign comes out negative, do not erase it. Interpret it: opposite direction, compression instead of tension, clockwise instead of counterclockwise, etc.

Final memory line - Equilibrium gives forces; compatibility gives indeterminate actions; unit load gives displacement; release gives influence line; horizontal thrust makes arches/cables efficient.